EXP.NO: 6	Write a Servlet to demonstrate session tracking using HttpSession. Implement a simple login system where the user's session is tracked.
DATE:24/02/2026	

AIM:

To demonstrate session tracking using HttpSession and implement a simple login system where the user's session is tracked.

ALGORITHM:

1. **Input:**
   - The user fills out a web form with their name and email.
   - The form will use both GET and POST methods for data submission.

2. **Steps for Creating and Handling the HTTP GET and POST Requests:**
   1. **Create an HTML Form:**
      - Design an HTML form with two input fields: one for the name and one for the email.
      - Create two buttons to allow the user to submit the form using GET or POST methods.
      - Use the action attribute to define the Servlet URL and specify the form method (either GET or POST).
   2. **Create the Servlet to Handle GET and POST Requests:**
      - Write a Java Servlet class that will handle both GET and POST requests.
      - The servlet will have two methods:
        - doGet() to handle the GET requests.
        - doPost() to handle the POST requests.

- In both methods, retrieve the form data using request.getParameter().
- In the doGet() method, process and display the data in the URL.
- In the doPost() method, process and display the data in the HTTP request body.

3. **Handle GET Requests in the Servlet:**
   - In the doGet() method, use request.getParameter() to retrieve the name and email submitted via GET.
   - Output the data as a response, including the name and email in the URL (visible to the user).
4. **Handle POST Requests in the Servlet:**
   - In the doPost() method, use request.getParameter() to retrieve the name and email submitted via POST.
   - Output the data as a response, displaying the name and email without exposing them in the URL (more secure).
5. **Return the Output:**
   - The Servlet should return an HTML page displaying the name and email entered by the user.
   - For the GET method, the output should be visible in the URL.
   - For the POST method, the output should be visible in the response body, not the URL.
6. **Deploy and Test the Servlet:**
   - Compile the Servlet class and deploy it to a Servlet container (like Apache Tomcat).
   - Test the form using GET and POST methods to see the difference in data submission.
7. **Observe and Compare GET and POST Results:**

- Test the GET method: The form data is sent in the URL.
- Test the POST method: The form data is sent in the HTTP request body, not visible in the URL.

PROGRAM:

Exp6.html:

```
<!DOCTYPE html>
<html>
<head>
    <title>Library Login</title>

    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.
min.css" rel="stylesheet">

    <style>
        body {
            background: linear-gradient(to right, #eef2f3, #dfe9f3);
        }
    </style>
</head>

<body>

<div class="container d-flex justify-content-center align-items-center vh-
100">

    <div class="card p-4 shadow-lg border-0" style="width: 360px;">

        <h3 class="text-center mb-3"> Library Login</h3>
```

```
        <form method="post" action="LibraryServlet">

            <div class="mb-3">
                <label class="form-label">Username</label>
                <input type="text" name="username" class="form-control"
placeholder="Enter username" required>
            </div>

            <div class="mb-3">
                <label class="form-label">Password</label>
                <input type="password" name="password" class="form-control"
placeholder="Enter password" required>
            </div>

            <button type="submit" class="btn btn-primary w-
100">Login</button>
        </form>

    </div>

</div>

</body>
</html>

<!DOCTYPE html>
<html>
<head>
    <title>Dashboard</title>

    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.
min.css" rel="stylesheet">
```

```
    <style>
        body {
            background-color: #f8f9fa;
        }
    </style>
</head>

<body>

<nav class="navbar navbar-light bg-white shadow-sm px-3">
    <span class="navbar-brand fw-bold"> Library System</span>
    <span>
        Welcome, <strong><%= session.getAttribute("username") %></strong>
    </span>
    <a href="LibraryServlet?action=logout" class="btn btn-outline-danger
btn-sm">Logout</a>
</nav>

<div class="container mt-4">

    <h3 class="mb-4">Available Books</h3>

    <div class="row g-3">

        <div class="col-md-4">
            <div class="card p-3 shadow-sm border-0">
                <h5>Java Programming</h5>
                <p class="text-muted">Core & Advanced Java concepts</p>
                <button class="btn btn-success">Issue Book</button>
            </div>
        </div>

        <div class="col-md-4">
```

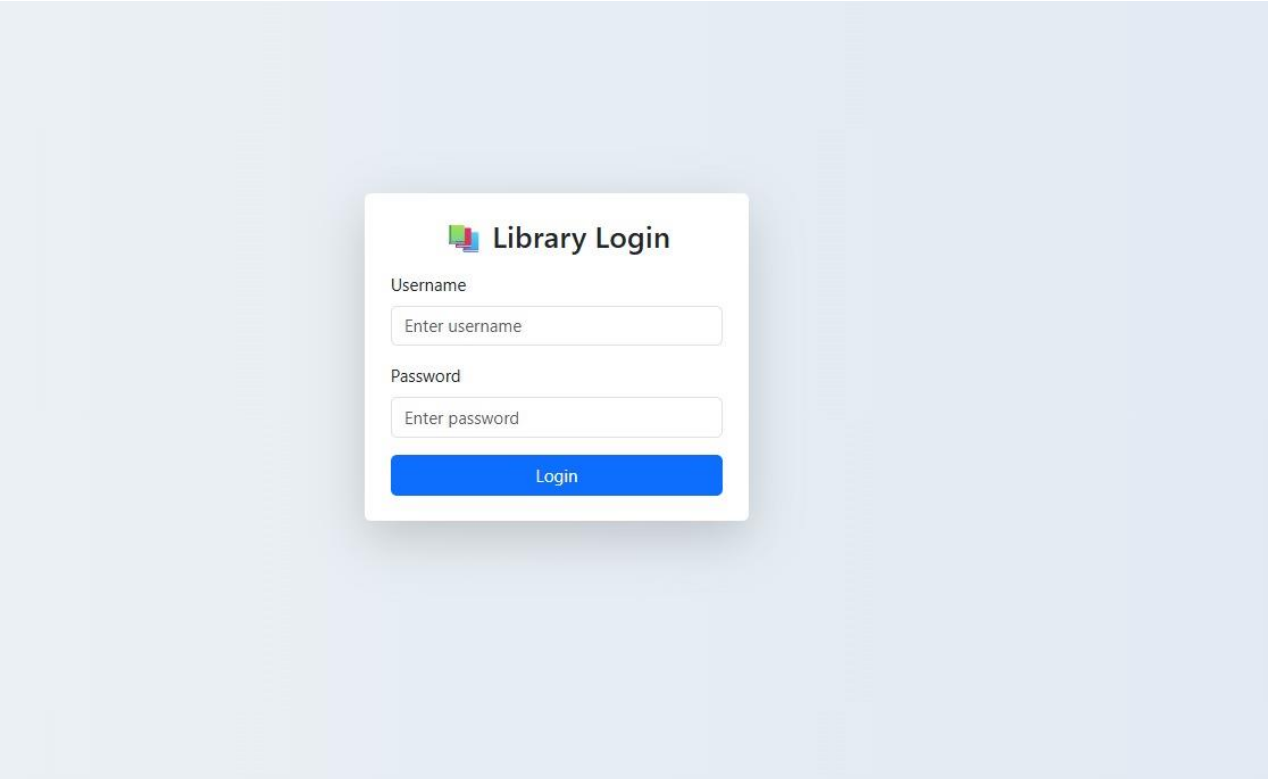
```
            <div class="card p-3 shadow-sm border-0">
                <h5>Data Structures</h5>
                <p class="text-muted">Prepare for placements</p>
                <button class="btn btn-success">Issue Book</button>
            </div>
        </div>

        <div class="col-md-4">
            <div class="card p-3 shadow-sm border-0">
                <h5>Web Development</h5>
                <p class="text-muted">Frontend & Backend basics</p>
                <button class="btn btn-success">Issue Book</button>
            </div>
        </div>

    </div>

</div>

</body>
</html>
```

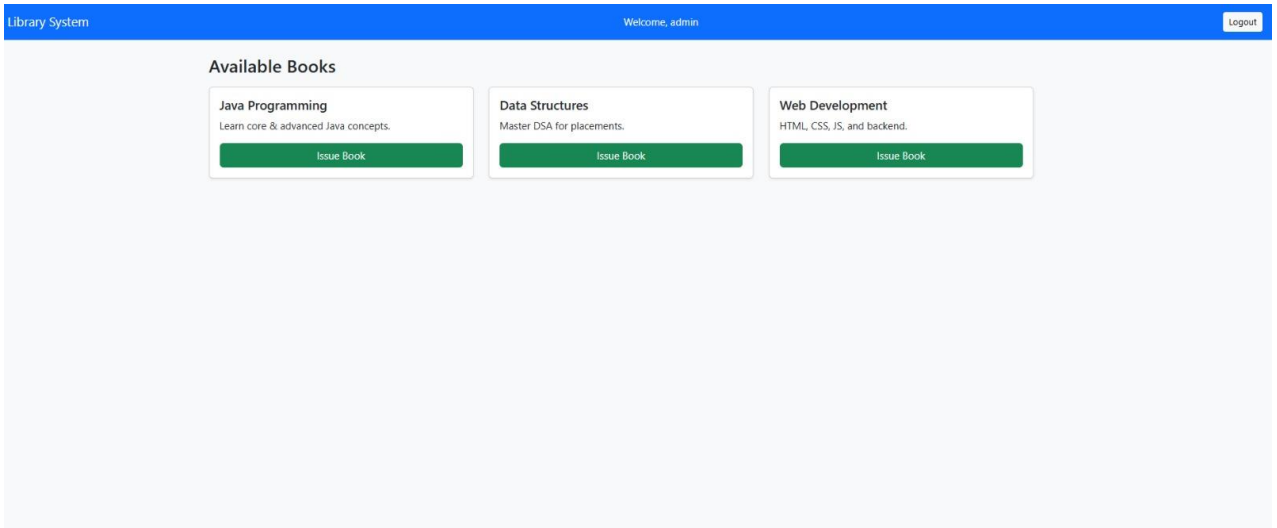
Servlet6.java:

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/LibraryServlet")
public class LibraryServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
```

```
        String action = request.getParameter("action");

        // Logout
        if ("logout".equals(action)) {
            HttpSession session = request.getSession(false);
            if (session != null) {
                session.invalidate();
            }
            response.sendRedirect("login.jsp");
            return;
        }

        // Check session
        HttpSession session = request.getSession(false);

        if (session != null && session.getAttribute("username") != null) {
            request.getRequestDispatcher("dashboard.jsp").forward(request,
response);
        } else {
            response.sendRedirect("login.jsp");
        }
    }

    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        String username = request.getParameter("username");
        String password = request.getParameter("password");

        // Hardcoded login
        if ("admin".equals(username) && "1234".equals(password)) {

            HttpSession session = request.getSession();
            session.setAttribute("username", username);
            session.setMaxInactiveInterval(60); // 60 sec

            response.sendRedirect("LibraryServlet");

        } else {
            response.sendRedirect("error.jsp");
        }
    }
}
```

Output:


Library Login
Username
Enter username
Password
Enter password
Login


Library System
Welcome, admin
Logout
Available Books
Java Programming
Learn core & advanced Java concepts.
Issue Book
Data Structures
Master DSA for placements.
Issue Book
Web Development
HTML, CSS, JS, and backend.
Issue Book

RESULT:

A Servlet is created to demonstrate session tracking using HttpSession and implemented a simple login system where the user's session is tracked.